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SOLAR & STORAGE PROPOSAL

MAISON M · NACCACHE

Solar system with battery storage to cut the generator hours — four options, sized on your measured load, with the generator carrying the whole site (no EDL)

PREPARED FOR
Mr. Makhlouf
Maison M, Naccache

DATE
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PREPARED BY
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REFERENCE
SG-2026-MKH-01 rev 2 · valid 30 days

1. Your site today

power logger on the generator, 3 – 8 October 2026

We installed a three-phase power logger on the generator busbar and recorded every minute for five days. The figures below are measured, not assumed; the yearly load is set to the 75,000 kWh a month you gave us at our meeting of 1 October. **Revision 2:** as EDL is becoming expensive, this revision assumes the generator carries the whole load, 24 hours a day — the ≈ 4.8 hours a day that EDL covered during the logging are now on the generator too.

AVERAGE LOAD
104 kW
≈ 2,490 kWh a day

PEAK DEMAND
192 kW
3 Oct 18:53 · 376 A on L1

GENERATOR RUNNING
24 h/day
logged 19.2 h + the 4.8 h EDL covered

DIESEL ON THE GENERATOR ALONE
≈ 830 L/day
≈ 302,200 L a year ≈ $302,200 at $1/L (700 L/day with EDL)

Generators: G1 350 kVA and G2 600 kVA on a manual changeover, ATS to EDL. Load profile: 70–90 kW at night, 130–190 kW in the evening. Yearly basis: 900,000 kWh (75,000 kWh/month), all of it from the generator. Diesel calibrated on the 700 L/day you reported for the logged days (19.2 h on the generator), which gives ≈ 3 kWh per litre on your generators; on the generator alone the same site burns ≈ 830 L/day.

2. The four options

same site, same year of sun, same load — only the system changes

Option	System	kWh saved / year	Solar share of load	Generator	Diesel saved / year	Price (excl. VAT)	Payback
A	292 panels · 188.3 kWp on-grid · no battery	154,914	17 %	24 h/day	21,263 L 7 % · $21,300	**$79,000**	3.7 yr
B	292 panels · 188.3 kWp + Deye 768 kWh battery · 375 kW	288,324	32 %	19.2 h/day	73,549 L 24 % · $73,500	**$229,000**	3.1 yr
C	600 panels · 387 kWp on-grid · no battery	171,572	19 %	24 h/day	23,398 L 8 % · $23,400	**$162,330**	6.9 yr
D RECOMMENDED	600 panels · 387 kWp + Deye 768 kWh battery · 375 kW	**513,237**	**57 %**	**13.7 h/day**	**146,977 L** 49 % · $147,000	**$312,330**	**2.1 yr**
	No solar (today)	**0**	**0 %**	**24 h/day**	**≈ 302,182 L of diesel a year ≈ $302,200**		—

Prices are all-in: panels, mounting structures, MPPT / PCS / STS equipment, cabling, protection boards, installation and commissioning. B and D add the battery clusters and the extra PCS modules to A and C. VAT 11 % on top — see section 5. kWh saved = solar energy the site uses instead of generator energy. Payback = price ÷ diesel saved at $1.00 per litre, without maintenance, battery replacement or fuel-price changes.

Where the solar energy goes, per year

Option	Solar made	Solar used	Solar thrown away	Battery to load	Generator kWh	Generator kWh saved
A · 292 panels, on-grid	288,324	154,914	133,410 · 46 %	—	745,086	154,914 · 17 %
B · 292 panels + 768 kWh	288,324	288,324	0 · 0 %	56,603	619,275	280,725 · 31 %
C · 600 panels, on-grid	592,447	171,572	420,875 · 71 %	—	728,428	171,572 · 19 %
D · 600 panels + 768 kWh	**592,447**	**513,237**	79,210 · 13 %	175,346	**405,981**	**494,019** · 55 %
No solar (today)	**0**	**0**	—	—	**900,000**	—

All figures in kWh a year. Solar thrown away = energy the panels could make but the site cannot take, because the generator must stay above its minimum load and there is no battery to store it. Generator kWh saved is against the 900,000 kWh the generator supplies when it carries the whole load (no EDL).

Why the battery matters on a generator site. Without a battery (A and C) the generator must keep running at a minimum load, so solar can only shave the daytime load: 46 % (A) to 71 % (C) of the solar energy is thrown away and the generator still runs 24 hours a day. With the battery (B and D) almost nothing is wasted, the generator stops whenever solar and battery can carry the load, and the diesel saving goes from 7–8 % to 24–49 %. Without EDL the battery works harder: on option D it carries the load ≈ 10 hours a day.

3. What each option includes

Option A — $79,000

292 × 645 W panels · 188.3 kWp · on-grid, no battery

- 292 bifacial 645 W panels (Longi HiMO X10 class)
- Three-phase on-grid string inverters, synchronised to the generator and to EDL
- Fuel-save controller: keeps the generator above its minimum load, no reverse power
- Structures, cabling, boards, installation and commissioning

Option B — $229,000

292 × 645 W panels · 188.3 kWp · 768 kWh / 375 kW battery

- 292 bifacial 645 W panels
- Deye 1 × MPPT module 200 kWp
- Deye 3 × PCS 125 kW = 375 kW
- Deye 1 × STS 500 kW static switch
- Deye 3 × 256 kWh lithium battery clusters = 768 kWh, with racks and BMS
- Storage kit on top of A (+ $150,000): the string inverters are replaced by Deye MPPT + PCS + STS

Option C — $162,330

600 × 645 W panels · 387 kWp · on-grid, no battery

- 600 bifacial 645 W panels
- Three-phase on-grid string inverters, synchronised to the generator and to EDL
- Fuel-save controller: keeps the generator above its minimum load, no reverse power
- Structures, cabling, boards, installation and commissioning
- Shown for comparison only — see section 4

Option D — RECOMMENDED — $312,330

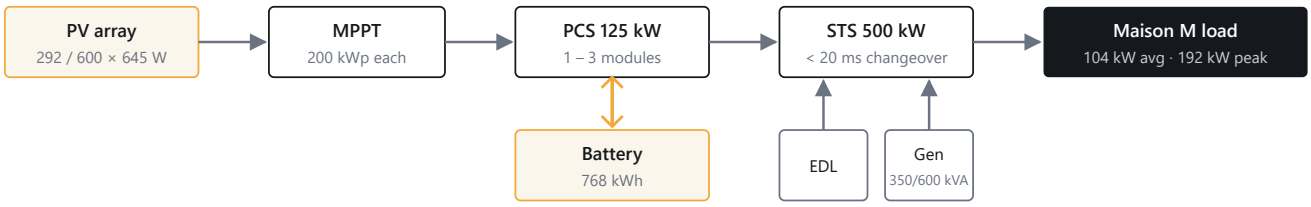
600 × 645 W panels · 387 kWp · 768 kWh / 375 kW battery

- 600 bifacial 645 W panels
- Deye 2 × MPPT module 200 kWp
- Deye 3 × PCS 125 kW = 375 kW
- Deye 1 × STS 500 kW static switch
- Deye 3 × 256 kWh lithium battery clusters = 768 kWh, with racks and BMS
- Storage kit on top of C (+ $150,000): the string inverters are replaced by Deye MPPT + PCS + STS

The storage system (options B and D)

Options A and C are classic on-grid systems: string inverters feed the solar into the busbar while the generator or EDL carries the rest. Options B and D are built on the Deye BOS-B Pro-A3 commercial energy-storage platform (inverter and battery both Deye) that takes over the generator and EDL changeover:

- **PCS 125 kW** modules, 98.5 % efficiency, 170 % peak power for motor starts; parallel up to 2.5 MW.
- **STS 500 kW** static switch: EDL, generator and load on three independent 500 kW paths, changeover in under 20 ms — the site never sees the switch.
- **MPPT 200 kWp** modules, 8 trackers each, > 99 % efficiency, 800 V PV.
- **Lithium battery** 256 kWh per cluster (16 × 16 kWh packs), independent BMS per cluster, IP20 racks for an indoor or sheltered room.
- **Built-in EMS**: generator auto start/stop on battery level, zero-export, time-of-use; colour touchscreen and cloud monitoring.
- PCS and MPPT modules are IP65 rated for outdoor mounting.



4. Our recommendation

Option D — 600 panels with the 768 kWh battery. It is the only option sized to the whole site: solar covers 57 % of the yearly load, the generator runs under 14 hours a day instead of 24, and diesel drops by about 147,000 litres a year. It pays for itself in about 2.1 years and the equipment is rated for well over ten. If the budget has to be staged, **Option B** is the smaller complete system at 3.1 years payback. We do not advise Option C: 600 panels without storage throw away 71 % of their energy and the generator still runs 24 hours a day, so the extra panels pay back in about 7 years.

5. Scope and exclusions

Included

- Engineering: load study, PV layout, single-line diagram, structural design of the mounting structures
- Supply of panels, inverters or MPPT / PCS / STS, and batteries as listed per option, delivered to site
- Steel mounting structures (roof and car-park carport, which doubles as covered parking)
- DC and AC cabling, combiner and protection boards, earthing, cable trays
- Installation, crane and commissioning labour
- Programming of generator start/stop, EDL changeover and battery limits; cloud monitoring
- Thermal-camera inspection at commissioning, labelling, operator training
- Manufacturer warranty on all equipment (terms to be confirmed with the order)

Not included

- VAT 11 % on all prices
- Civil works: a battery room or shelter if none exists
- Any change to the generators, the ATS or the EDL connection beyond the STS cabling
- Panels beyond the layout agreed; the roof / car-park split is fixed at the design stage

6. Basis of the figures

- Load: five days of one-minute logger data on the generator (3–8 Oct 2026), scaled to your 75,000 kWh a month. The hours EDL covered during the logging (≈ 4.8 h/day) are counted on the generator: the whole 900,000 kWh a year comes from the generator, none from EDL.
- Sun: a full year of five-minute production from a monitored plant in Naccache, corrected to the long-term PVGIS average for the site; 645 W panels, orientation to be fixed in the layout.
- Generator kept at a minimum of 20 % of its rating while running; battery working between 10 % and 100 %, generator restarts at 25 % and stops at 40 %.
- Diesel: the generator curves are calibrated on the 700 L/day you reported for the logged days (19.2 h on the generator); on the generator alone that is ≈ 830 L/day. Priced at $1.00 per litre — a higher diesel price shortens every payback.

7. Commercial terms

- Prices in US dollars, excluding VAT (11 %). Offer valid 30 days.
- Payment: 50 % with the order, 40 % on delivery of the equipment to site, 10 % at commissioning (to be confirmed).
- Delivery: 8 – 10 weeks from order confirmation for the storage equipment; panels and structures in parallel.

We would be glad to walk you through the options on site and fix the panel layout together.

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Accepted for Maison M
name · signature · date